

Assignment 2: Bandwidth Allocation Problem

ICT4SS - Operational Research Assignement 2

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1 Problem Description

Given an undirected graph $G = (V, E)$, where nodes V correspond to Tx-Rx pairs in a wireless environment and edges E represent potential conflicts[cite: 8]. We must assign a frequency $f_i \in \{1, \dots, k\}$ to every node ensuring that adjacent nodes use neither the same frequency nor contiguous frequencies ($|f_i - f_j| \geq 2$)[cite: 8]. The goal is to minimize the total width of the used spectrum[cite: 9].

2 Integer Linear Programming (ILP) Formulation

2.1 Decision Variables

- $x_{if} \in \{0, 1\}$: 1 if node $i \in V$ is assigned to frequency $f \in \{1, \dots, k\}$, 0 otherwise.
- $y_f \in \{0, 1\}$: 1 if frequency f is used by at least one node, 0 otherwise.

2.2 Objective Function

Minimize the total number of frequency slots used:

$$\min \sum_{f=1}^k y_f \tag{1}$$

2.3 Constraints

- **Unique Assignment:** Each node must be assigned exactly one frequency:

$$\sum_{f=1}^k x_{if} = 1, \quad \forall i \in V \tag{2}$$

- **Frequency Activation:** A frequency f is marked as used if any node is assigned to it:

$$x_{if} \leq y_f, \quad \forall i \in V, \forall f \in \{1, \dots, k\} \quad (3)$$

- **Conflict and Contiguity Constraints:** For every edge $(i, j) \in E$, adjacent nodes cannot use frequencies $f, f-1, f+1$:

- For $f = 1$:

$$x_{i,1} + x_{j,1} + x_{j,2} \leq 1 \quad (4)$$

- For $1 < f < k$:

$$x_{if} + x_{j,f-1} + x_{j,f} + x_{j,f+1} \leq 1 \quad (5)$$

- For $f = k$:

$$x_{ik} + x_{j,k-1} + x_{j,k} \leq 1 \quad (6)$$

- **Spectrum Compactness :** To avoid gaps and improve solver performance:

$$y_f \geq y_{f+1}, \quad \forall f \in \{1, \dots, k-1\} \quad (7)$$

3 Relationship with Graph Coloring

The Bandwidth Allocation (BA) problem is a generalization of the classic Graph Coloring problem[cite: 11, 12].

3.1 Theoretical Derivation

In standard coloring, the constraint for $(i, j) \in E$ is $|c_i - c_j| \geq 1$. In BA, the constraint is $|f_i - f_j| \geq 2$. We can establish a relationship using a linear mapping from an optimal coloring $c_i \in \{1, \dots, \chi(G)\}$.

By setting:

$$f_i = 2c_i - 1 \quad (8)$$

We observe that for any two adjacent nodes i, j :

$$|f_i - f_j| = |(2c_i - 1) - (2c_j - 1)| = 2|c_i - c_j| \quad (9)$$

Since $|c_i - c_j| \geq 1$ in a valid coloring, then $|f_i - f_j| \geq 2$, satisfying the BA requirements.

3.2 Optimal Spectrum Width

The resulting optimal spectrum width W relates to the chromatic number $\chi(G)$ as follows:

$$W = 2\chi(G) - 1 \quad (10)$$